

Gabriel Rodrigues

Bioinformatician · Viral Genomics

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Remote · GMT-3

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TECHNICAL STACK

R & Bioconductor	5+ yrs
Bash & Linux / HPC	5 yrs
Python	3+ yrs
Quarto & R Markdown	4 yrs
R Shiny	2+ yrs
Nextflow	2 yrs

CORE EXPERTISE

- NGS and bulk RNA-Seq analysis
- Viral discovery from public data
- Metagenomics & metatranscriptomics
- Genome / transcriptome assembly
- Differential expression & GSEA
- Small RNA and EVE analysis

TOOLS

Git · Docker · pixi · Conda · uv · SLURM · AWS · ggplot2 · Plotly · D3.js · bslib · renv · SQL

EDUCATION

2026 — Present
BSc, Computer Science
Centro Universitário Internacional (UNINTER)
Data science & algorithm / pipeline optimization

2025 — Present
PhD, Genetics and Molecular Biology
Universidade Estadual de Santa Cruz
Next-generation ViralQuest

Profile

Bioinformatician working where virology meets genomics. I design and maintain reproducible pipelines that turn public sequencing data into biological insight, from raw NGS reads to curated viral genomes and interactive reports. My work spans viral discovery and surveillance, metatranscriptomics, small RNAs and plant immunity, built on R, Python, Bash and Nextflow running on Linux HPC (SLURM) and cloud environments. Currently also pursuing a BSc in Computer Science, focused on data science and the optimization of algorithms and pipelines. I care about open, well-documented software: the viruses are already in the data, and my job is to make finding them routine.

Experience

PhD Researcher — Bioinformatics & Viral Genomics
2025 — Present
Biotechnology and Genetics Center, UESC · Ilhéus, BA
- Lead the full refactor of **ViralQuest** (Python ≥ 3.12, pixi), improving dependency installation, Apple silicon support and the interactive HTML report.
- Build automated NGS workflows in Python, Bash and Nextflow for Linux HPC (SLURM) and AWS.
- Extend viral discovery to new hosts and datasets mined from public sequence repositories.

Public Health Data Analyst & R Shiny Developer
2025 — 2026
UESC · Regional Health Authorities · Bahia
- Developed R Shiny dashboards that integrated disease notification data (SINAN) for public health teams.
- Automated reporting pipelines; deployed with bslib, shinymanager, renv and Docker for secure, reproducible releases.

MSc Researcher — Viral Genomics & Transcriptomics
2023 — 2025
Biotechnology and Genetics Center, UESC · Ilhéus, BA
- Mined public RNA-Seq libraries of *Theobroma cacao* for viral discovery and genomic surveillance.
- Performed bulk RNA-Seq analysis (assembly, differential expression, enrichment) with R/Bioconductor.
- Created the first release of ViralQuest and co-authored studies on viromes, EVEs and bee miRNAs.

Selected Projects

ViralQuest
Python · PyPI
Creator & lead developer
Interactive, accessible pipeline for viral sequence discovery, annotation and curation, with a D3.js HTML report for end users. Published in *BMC Bioinformatics* (2026).
Python · Bioinformatics · NGS Pipelines · Data Visualization · D3.js

viralfetch & slin
Python · TUI
Open-source command-line tools
viralfetch queries and downloads viral taxonomy, metadata and sequences; **slin** is a terminal interface for the SLURM workload manager.
Python · CLI & TUI · SLURM · HPC · Data Retrieval

Open science & technical writing
R · Quarto
gabrielvpina.github.io · dataViz · quartoTemplates
Personal blog with hands-on tutorials (Sanger sequencing to phylogenetic trees, decision trees and exploratory data analysis in R), plus open figure code from published papers and community Quarto templates, including this CV.
Tutorials · Data Science · Machine Learning · Phylogenetics · Quarto

Reproducible science templates
R · Quarto
dataViz · quartoTemplates
Open figure code from published papers and community Quarto templates, including this CV.
R · Quarto · Typst · ggplot2 · Reproducibility

2023 — 2025
MSc, Genetics and Molecular Biology
Universidade Estadual de Santa Cruz
Viral surveillance of Theobroma cacao

2018 — 2022
BSc, Interdisciplinary Sciences
Universidade Federal do Sul da Bahia

LANGUAGES

Portuguese	Native
English	Professional
Spanish	Intermediate

Selected Publications

2026 **ViralQuest: a user-friendly interactive pipeline for viral-sequences analysis**
Rodrigues, G. V. P., Ferreira, L. Y. M., Aguiar, E. R. G. R.
BMC Bioinformatics · doi:10.1186/s12859-026-06391-6

2026 **Tracing ancient viral footprints: endogenous viral elements in *Bombus* species**
Conceição, L. B. A., Santos, J. P. N., Ferreira, L. Y. M., **Rodrigues, G. V. P.**, et al.
Mobile DNA · doi:10.1186/s13100-026-00393-0

2025 **Insights into miRNAs of the stingless bee *Melipona quadrifasciata***
Soares, D. O., Ferreira, L. Y. M., **Rodrigues, G. V. P.**, et al.
Non-Coding RNA · doi:10.3390/ncrna11030048

2025 ***Theobroma cacao* virome: exploring public RNA-Seq data for viral discovery and surveillance**
Rodrigues, G. V. P., Santos, J. P. N., Ferreira, L. Y. M., et al.
Viruses · doi:10.3390/v17050624

2024 **The virome of cocoa fermentation-associated microorganisms**
Santos, J. P. N., **Rodrigues, G. V. P.**, Ferreira, L. Y. M., et al.
Viruses · doi:10.3390/v16081226